A Finite-Difference Solver for the 2D Neutron Diffusion Equation

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1 Introduction

Neutron diffusion models play a central role in reactor physics by approximating neutron transport in multiplying media. In this project, we develop a two-dimensional finite-difference solver for the steady-state neutron diffusion equation with spatially varying material properties. Both fixed-source and criticality formulations are considered, with numerical solutions obtained via sparse linear algebra methods.

2 Governing Equations

2.1 Fixed-Source Diffusion

The steady-state neutron diffusion equation in two dimensions is given by

$$-\nabla \cdot (D(x,y)\nabla\phi(x,y)) + \Sigma_a(x,y)\phi(x,y) = S(x,y), \tag{1}$$

where:

- $\phi(x,y)$ is the neutron flux,
- $D(x,y)$ is the diffusion coefficient,
- $\Sigma_a(x,y)$ is the absorption cross section,
- $S(x,y)$ is an external source.

Dirichlet boundary conditions $\phi = 0$ are imposed on the domain boundary.

2.2 Criticality Eigenvalue Problem

To model a multiplying system, we consider the one-group criticality equation:

$$-\nabla \cdot (D\nabla\phi) + \Sigma_a\phi = \frac{1}{k}\nu\Sigma_f\phi, \tag{2}$$

where $\nu\Sigma_f$ represents neutron production via fission and k is the effective multiplication factor. This can be written as a generalized eigenvalue problem:

$$A\phi = \lambda F\phi, \quad \lambda = \frac{1}{k}. \tag{3}$$

3 Numerical Method

The domain is discretized using a uniform Cartesian grid. A five-point finite-difference stencil is used to approximate the Laplacian:

$$\nabla^2 \phi \approx \frac{\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j}}{h_x^2} + \frac{\phi_{i,j+1} - 2\phi_{i,j} + \phi_{i,j-1}}{h_y^2}. \tag{4}$$

For spatially varying diffusion coefficients, harmonic averaging is used at cell interfaces to preserve flux continuity.

The discretization leads to a sparse linear system:

$$A\Phi = b, \tag{5}$$

which is solved using SciPy's sparse direct solvers. The criticality problem is solved using a sparse generalized eigenvalue solver.

4 Material Configurations

Several heterogeneous configurations were considered:

- Regions with varying diffusion coefficients to represent different materials,
- Localized high-absorption regions modeling control rods,
- Fissile regions with nonzero $\nu\Sigma_f$ for criticality analysis.

5 Results

5.1 Fixed-Source Simulation

Figure 1 shows the neutron flux distribution for a heterogeneous medium with a localized source and absorber region.

The absorber region significantly reduces the local flux, while higher diffusion regions smooth the flux profile.

5.2 Grid Refinement Study

To assess numerical behavior, simulations were performed on increasingly refined grids.

As the grid is refined, the peak flux approaches a stable value, indicating convergence of the numerical solution.

5.3 Criticality Analysis

The dominant eigenmode and corresponding effective multiplication factor were computed.

The computed value of k_{eff} indicates whether the system is subcritical ($k < 1$), critical ($k = 1$), or supercritical ($k > 1$). Spatial variations in absorption and fission properties strongly influence the flux shape.

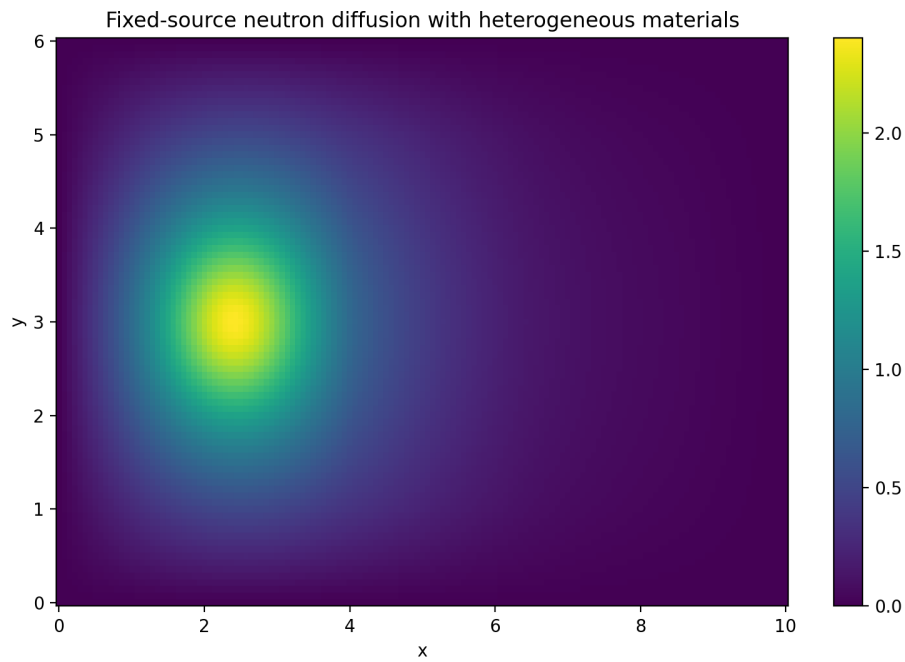

Figure 1: Neutron flux for a heterogeneous medium with source and absorber regions.

6 Discussion

The results demonstrate the expected physical behavior:

- Absorber regions suppress neutron flux locally.
- Higher diffusion coefficients lead to smoother spatial distributions.
- The eigenvalue solution reflects the balance between neutron production and loss.

The grid refinement study suggests that the finite-difference method provides consistent approximations as the mesh is refined.

7 Conclusion

A two-dimensional neutron diffusion solver was developed using finite-difference discretization and sparse linear algebra techniques. The implementation supports heterogeneous material properties and both fixed-source and eigenvalue formulations. The results align with expected physical behavior and demonstrate the capability of the model to capture key aspects of neutron transport.

8 Future Work

Potential extensions include:

- Implementation of vacuum or Robin boundary conditions,
- Multigroup diffusion models,

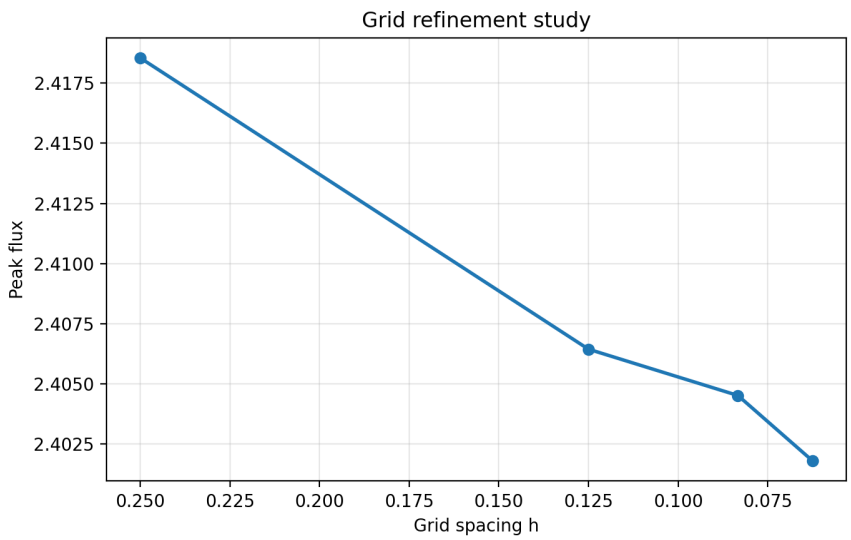

Figure 2: Peak flux versus grid spacing.

- Time-dependent neutron diffusion,
- Comparison with finite-element methods.

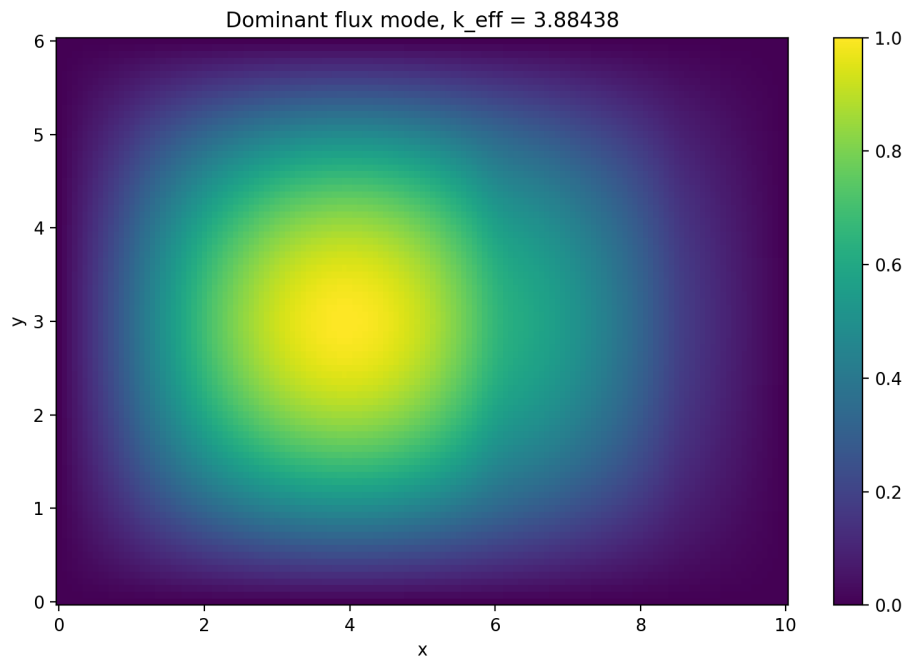

Figure 3: Dominant neutron flux mode for the criticality problem.